Important Information

Thank you for choosing Freenove products!

Getting Started

If you have not yet downloaded the zip file associated with this kit, please do so now and unzip it.
https://github.com/Freenove/Freenove_Tower_Cooler_for_Raspberry_Pi

Get Support and Offer Input

Freenove provides free and responsive product and technical support, including but not limited to:

- Product quality issues
- Product use and build issues
- Questions regarding the technology employed in our products for learning and education
- Your input and opinions are always welcome
- We also encourage your ideas and suggestions for new products and product improvements

For any of the above, you may send us an email to:

support@freenove.com

Safety and Precautions

Please follow the following safety precautions when using or storing this product:

- Keep this product out of the reach of children under 6 years old.
- This product should be **used only when there is adult supervision present** as young children lack necessary judgment regarding safety and the consequences of product misuse.
- This product contains small parts and some parts that are sharp. This product contains electrically conductive parts. **Use caution when placing conductive parts near power supplies, batteries, or live circuits.**
- When the product is powered on, some parts will move or rotate. **To avoid injury, keep your hands and fingers away from moving parts.**
- It is possible that an improperly connected or shorted circuit may cause overheating. **If this happens, immediately disconnect the power supply or remove the batteries. Do not touch anything until the product has cooled down!** When everything is safe and cool, review the product tutorial to identify the cause.
- Only operate the product in accordance with the instructions and guidelines of this tutorial, otherwise parts may be damaged or you could be injured.
- Store the product in a cool dry place and avoid exposing the product to direct sunlight.
- After use, always turn the power OFF and remove or unplug the batteries before storing.

About Freenove

Freenove provides open source electronic products and services worldwide.

Freenove is committed to assist customers in their education of robotics, programming and electronic circuits so that they may transform their creative ideas into prototypes and new and innovative products. To this end, our services include but are not limited to:

- Educational and Entertaining Project Kits for Robots, Smart Cars and Drones
- Educational Kits to Learn Robotic Software Systems for Arduino, Raspberry Pi and micro: bit
- Electronic Component Assortments, Electronic Modules and Specialized Tools
- **Product Development and Customization Services**

You can find more about Freenove and get our latest news and updates through our website:

http://www.freenove.com

sale@freenove.com

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Contents

Important Information 1
Contents 3
Preface 4
List 5
    Metal Parts 5
    Acrylic Parts 5
    Machinery Parts 6
    Electronic Parts 6
    Wires 7
    Tools 7
    Required but NOT Contained Parts 7
Chapter 1 Control Board 9
    1.1 Introduction to Control Board 9
Chapter 2 Assembly 10
    2.1 Assemble the Tower Cooler 10
Chapter 3 APP Control 13
    3.1 Boot Behavior & Environment Settings 13
What's next? 19

Preface

Welcome to use the Freenove Tower Cooler for Raspberry Pi



[FNK0113A]



[FNK0113B]

This product is exclusively designed for the RPi 5, a popular single-board computer (SBC). Follow this tutorial, you can provide better heat dissipation performance for your RPi 5.

The RPi 5 delivers highly powerful performance with a CPU frequency of up to 2.4GHz. However, this increased power consumption also leads to significant heat generation. Without an appropriate heatsink, thermal throttling may severely limit the Raspberry Pi 5's performance.

In this context, our case kit came into being. It's able to provide perfect heat dissipation capability for your RPi 5, allowing your RPi 5 to deliver more powerful and stable performance.

If you encounter any issues, feel free to contact us for prompt and free technical support.

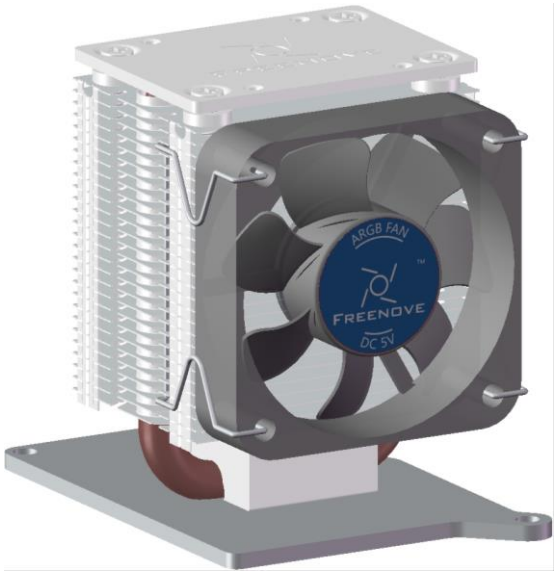
support@freenove.com

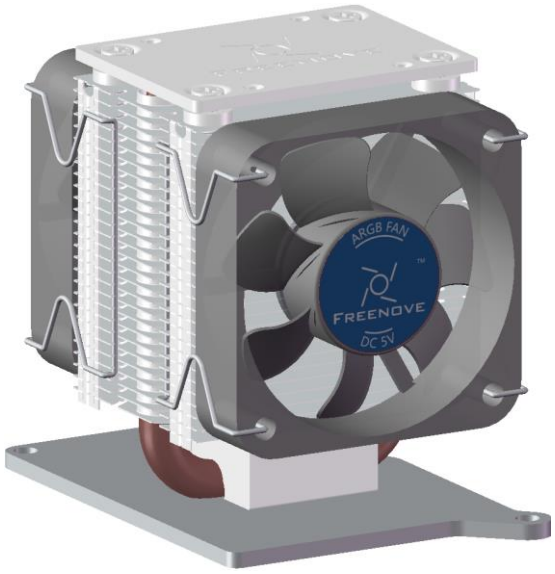
List

Before getting started, please check the part list. If any component is missing from your kit, do not start assembly; instead, please email our support team at support@freenove.com to get the missing parts.

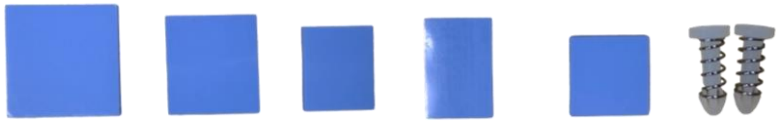
Metal Parts

Single-fan Tower Cooler x1 (left) or Dual-fan Tower Cooler x1 (right)





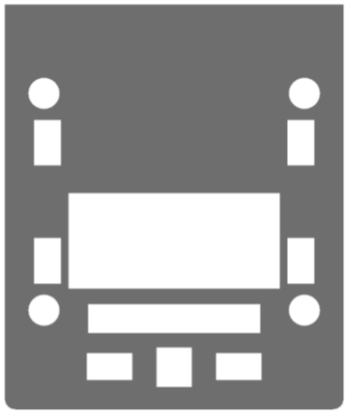
Thermal Pad x5, Nylon Standoff x2



Acrylic Parts

Note: Please peel off the protective films from the acrylic parts before use.

Acrylic x1



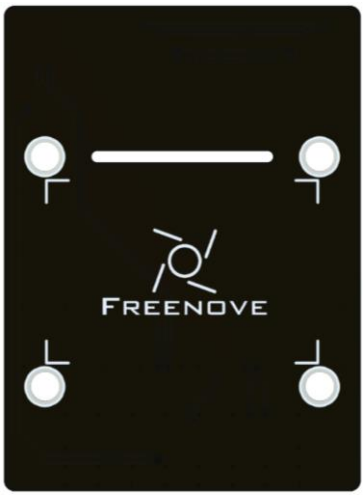
Machinery Parts

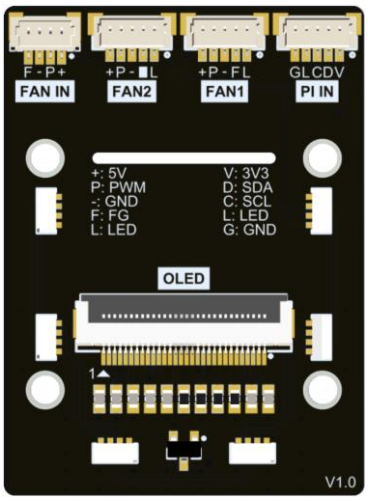


Electronic Parts

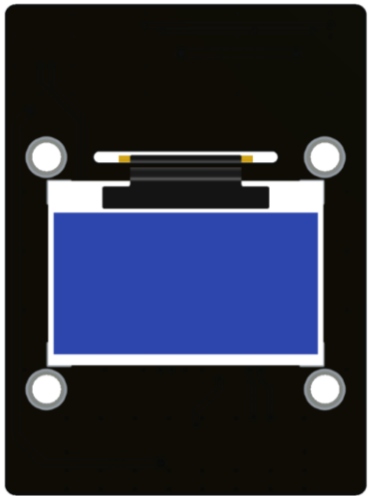
Control Board

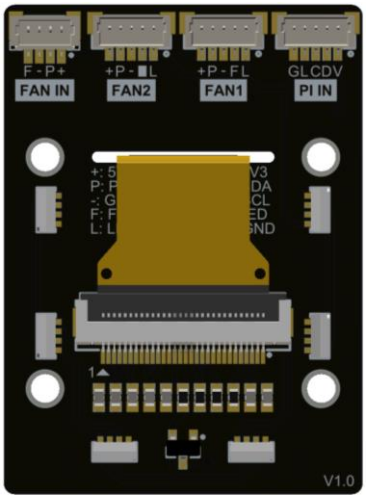
Control Board x1





Control Board x1(Only FNK0113B)





Wires

SH1.0mm_4P Same-Direction Cable 8cm x1

SH1.0mm_5P to 2.54mm Dupont Cable 7cm x1





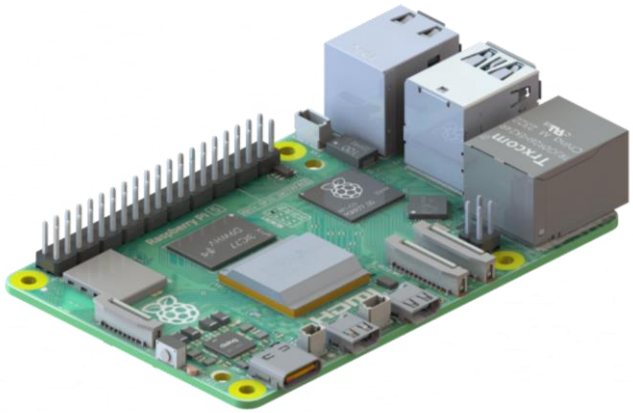
Tools

Cross screwdriver (3mm) x 1



Required but NOT Contained Parts

Raspberry Pi 5 x 1



27W Power Adapter x 1 (or a power adapter compatible with Raspberry Pi official one that can output 5.1V/5A)



Micro SD Card (TF Card), Card Reader x 1

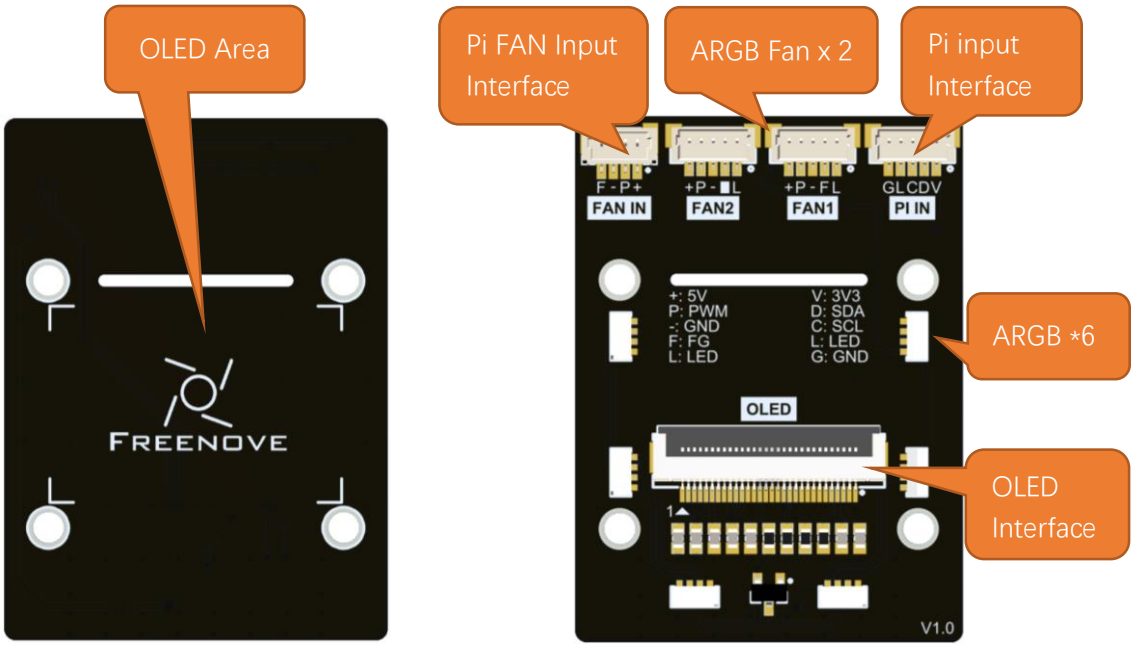


Before getting started, please check the part list. If any component is missing from your kit, do not start assembly; instead, please email our support team at support@freenove.com to get the missing parts.

Chapter 1 Control Board

In this chapter, we will introduce the functions of the control board.

1.1 Introduction to Control Board



This control board is equipped with multiple interfaces and features 6 built-in ARGB LEDs that can synchronize with ARGB fan lighting:

- Pi FAN Input Interface: RPi 5 fan control interface, controlling fan speed and providing speed feedback.
- ARGB FAN: Two channels of ARGB FAN (Note: FAN2 interface does not have speed feedback)
- Pi input Interface: Raspberry Pi IIC communication and LED control interface.
- OLED Interface: Connect to 0.96 inch OLED Screen, IIC communication protocol

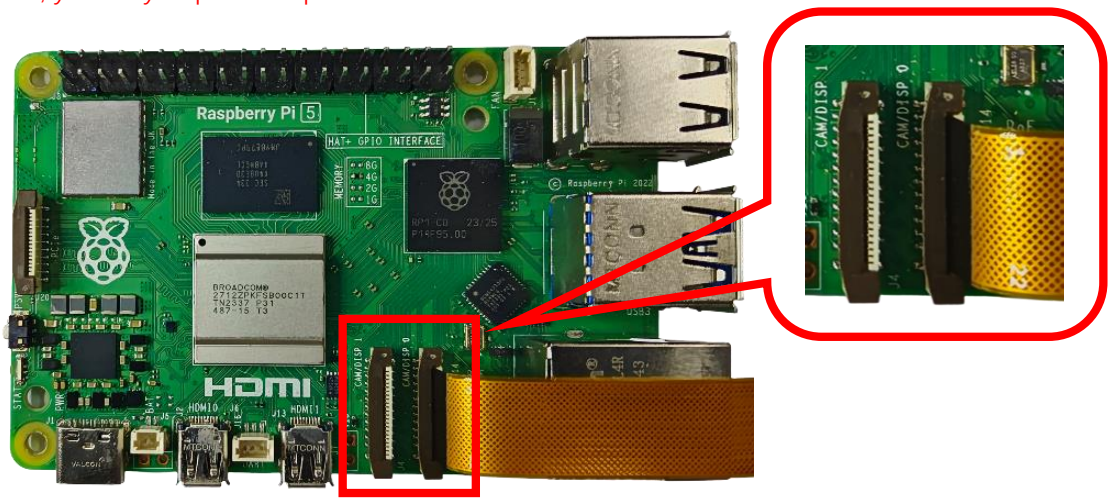
Chapter 2 Assembly

It is recommended to assemble and use the Freenove Tower Cooler for Raspberry Pi according to this tutorial. Otherwise, it may lead to incorrect device installation or damage. As noted earlier, please check the part list. If any component is missing from your kit, do not start assembly; instead, please email our support team at support@freenove.com to get the missing parts.

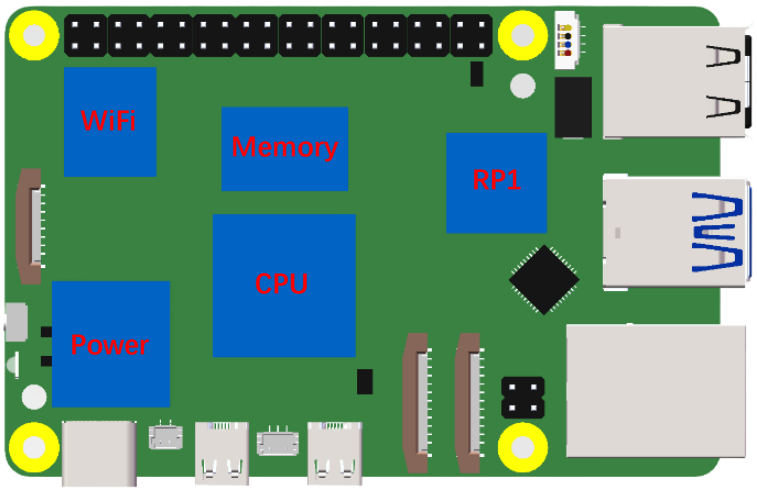
Make sure the device is powered off before assembly.

2.1 Assemble the Tower Cooler

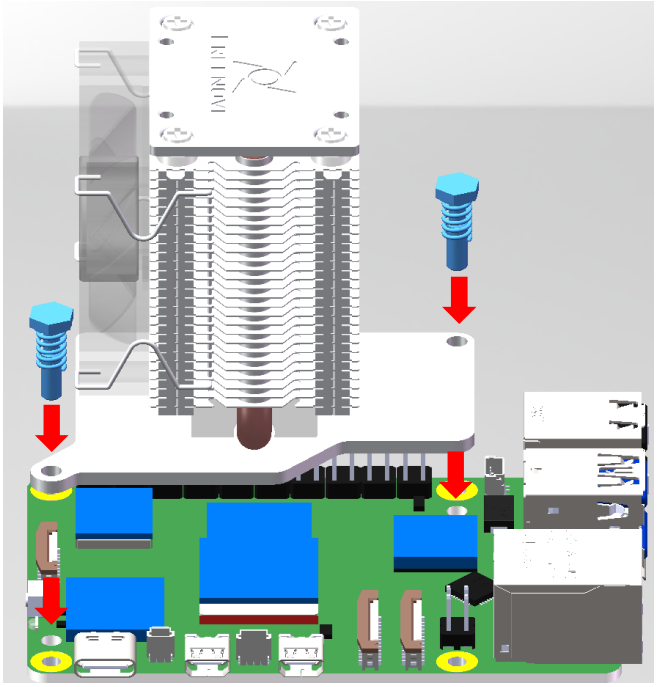
Note: The installation of a tower cooler may make it difficult to access the CAM/DISP interface. if you need to connect a device (such as a camera) to this interface, you must connect the cable before installing the cooler. Otherwise, you may skip this step.



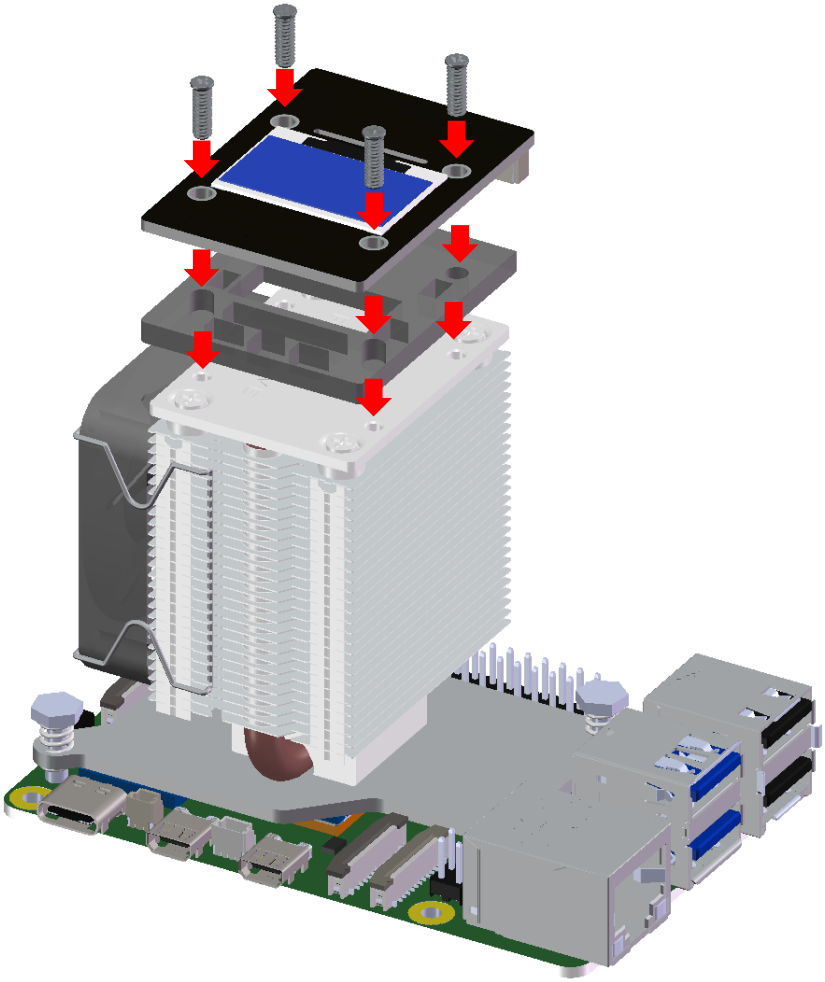
Step 1: Identify the five thermal pads by size and apply each one to its corresponding chip on the RPi 5. (Important: Remove the protective film from both sides of each thermal pad to ensure optimal thermal conductivity.)



Step 2: Fix the Tower Cooler to RPi 5 with the Nylon standoffs.



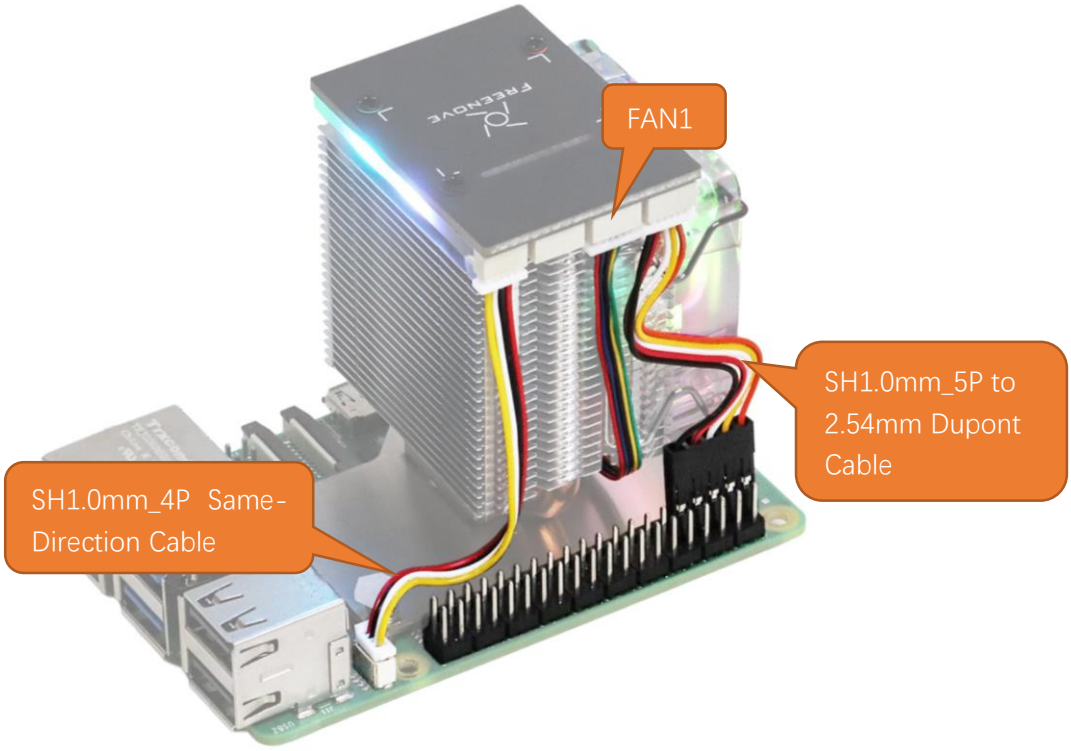
Step 3: Align the acrylic with the Control Board and secure it to the top of the RPi using M2.5*8 screws.



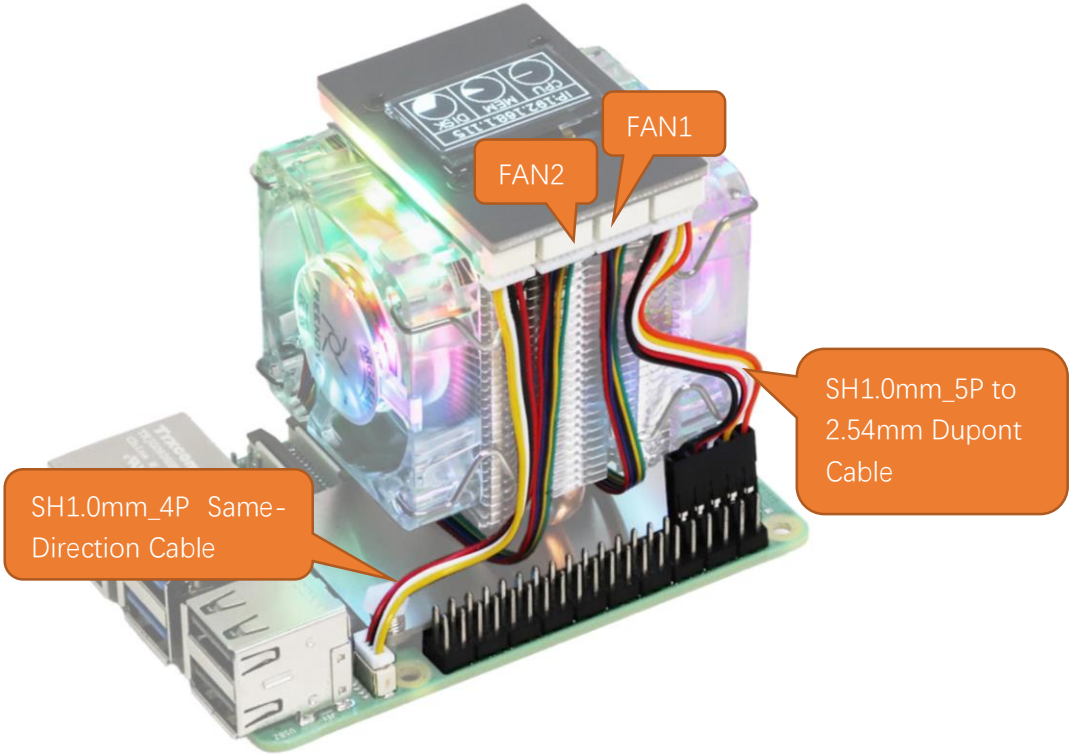
Step 4:

Notice: This connector is polarized (non-reversible). Forcing it or connecting it incorrectly will permanently damage the interface.

If you purchased FNK0113A, please wire according to the diagram below.



If you purchased FNK0113B, please wire according to the diagram below.



Chapter 3 APP Control

Before powering on the Freenove Tower Cooler for Raspberry Pi, please make sure that all cable connections are correct.

This case requires an adequate power supply. We highly recommend using the official Raspberry Pi 5.1V / 5A power adapter (https://www.raspberrypi.com/products/27w-power-supply).

Failure to do so may result in the Freenove Tower Cooler for Raspberry Pi being unusable or causing damage to components.

3.1 Boot Behavior & Environment Settings

3.1.1 What to Expect on First Startup

After installation is complete, please follow the steps below to confirm all functions are working properly.



1. When the RPi starts up, the fan should automatically spin twice, indicating the fan is running normally.

Troubleshooting Guide:

If the fan does not spin at startup, please: Immediately power off the system， and check if the SH1.0mm_4P Same-Direction Cable and SH1.0mm_5P to 2.54mm Dupont Cable are connected correctly.

If you have any questions of the above, please contact us at **support@freenove.com**

2. RPi 5 Status LED: The green STAT LED will remain steadily illuminated. If this LED is not lit or displays any pattern other than a steady green light, it indicates that the Raspberry Pi operating system has not booted successfully. In this case, please check your Raspberry Pi hardware and OS installation separately.



In this case, you can connect a screen to your Raspberry Pi or access it via VNC viewer.
If you have any questions of the above, please contact us at **support@freenove.com**

3.1.2 Software Setup

Software Packages Update

Run the following command on the terminal to update your Raspberry Pi's package list to the latest version. In the latest Raspberry Pi 13 (Trixie), passwordless sudo is officially disabled. You will need to enter your user password every time you run a sudo command. Please enter your user password when prompted. (Note: The default password is "**raspberry**", please replace it with your user password.)

```
sudo apt update
```

```
pi@raspberrypi:~ $ sudo apt update
[sudo] password for pi:
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://archive.raspberrypi.com/debian trixie InRelease
Hit:4 http://deb.debian.org/debian-security trixie-security InRelease
127 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Code downloading

Open the Raspberry Pi Terminal, type the following two commands one by one to download the code.

```
cd
git clone
https://github.com/Freenove/Freenove_Tower_Cooler_for_Raspberry_Pi.git
```

```
pi@raspberrypi:~ $ cd
pi@raspberrypi:~ $ git clone https://github.com/Freenove/Freenove_Tower_Cooler_for_Raspberry_Pi.git
Cloning into 'Freenove_Tower_Cooler_for_Raspberry_Pi'...
remote: Enumerating objects: 49, done.
remote: Counting objects: 100% (49/49), done.
remote: Compressing objects: 100% (44/44), done.
remote: Total 49 (delta 2), reused 45 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (49/49), 13.78 MiB | 1.63 MiB/s, done.
Resolving deltas: 100% (2/2), done.
```

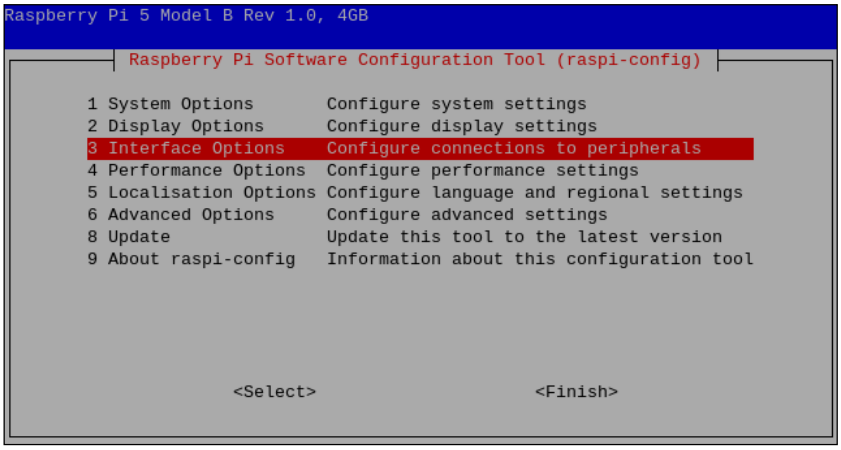
Enable I2C (Important)

The I2C function must be enabled, as it is required for communication between the Raspberry Pi, the OLED display. Without it, the chassis software will fail to operate.

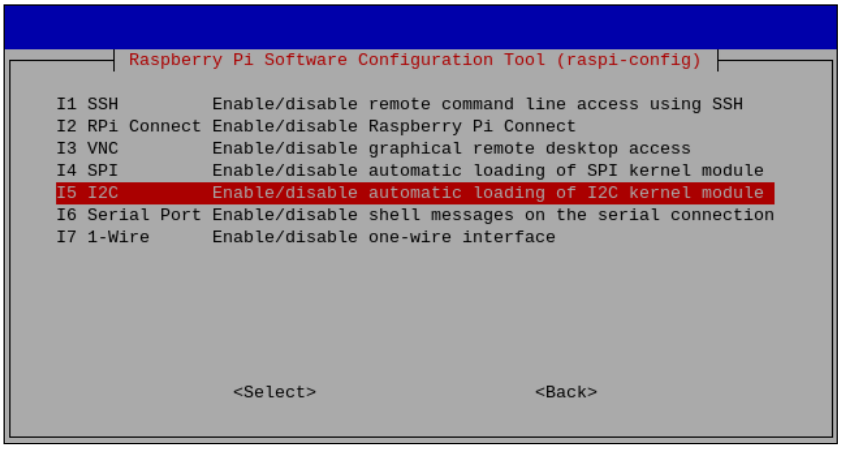
Run the following command to open the Raspberry Pi configuration interface.

```
sudo raspi-config
```

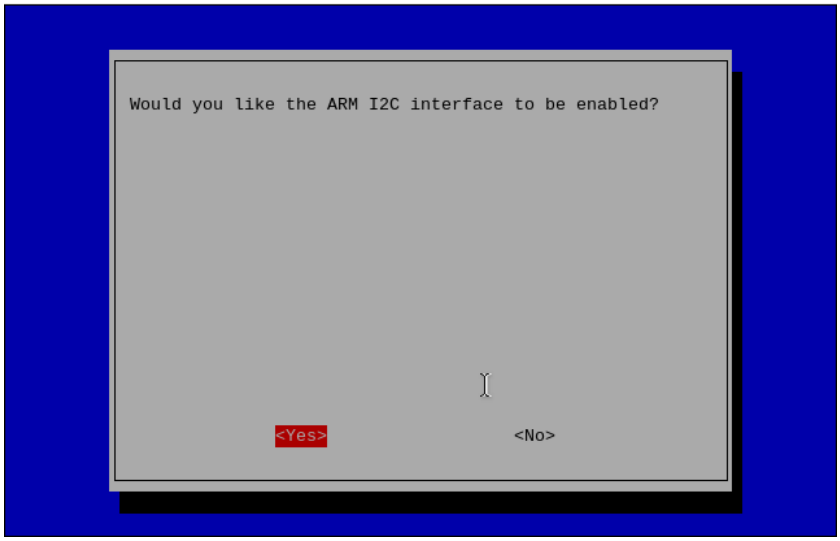
Navigate to Interface Options and press Enter.



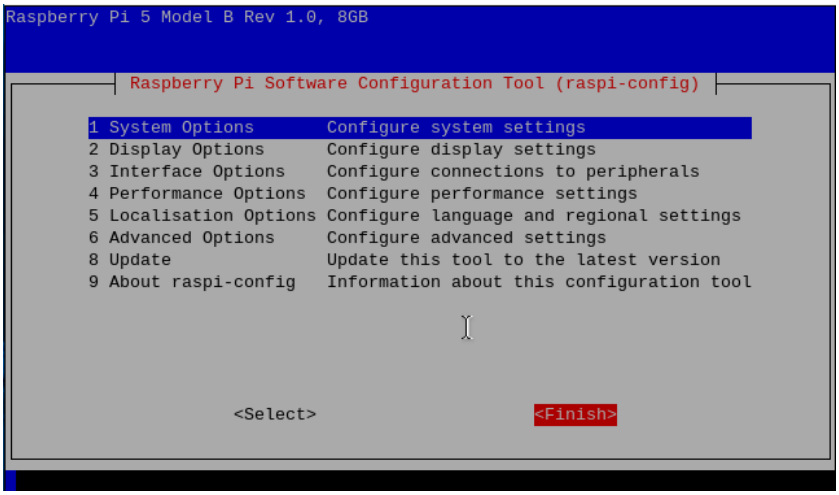
Select I2C and press Enter to enable it.



Select Yes, and press Enter.



Select Finish to save the change and exit the configuration interface.



Creating Desktop Shortcut

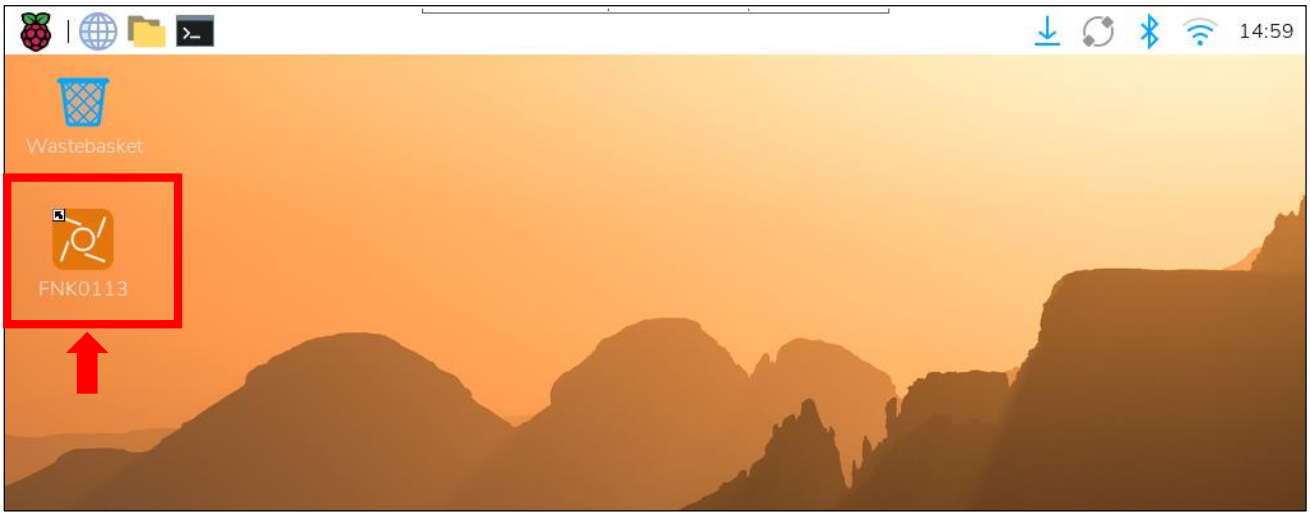
Run the following two commands one by one in the Raspberry Pi terminal to create a desktop shortcut for the case control software. Configuring the software environment takes some time, please be patient.

```
cd Freenove_Tower_Cooler_for_Raspberry_Pi
sudo python setup.py
```

```
pi@raspberrypi:~ $ cd Freenove_Tower_Cooler_for_Raspberry_Pi/
pi@raspberrypi:~/Freenove_Tower_Cooler_for_Raspberry_Pi $ sudo python setup.py
Starting setup...
Running: sudo apt update
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://archive.raspberrypi.com/debian trixie InRelease
Hit:4 http://deb.debian.org/debian-security trixie-security InRelease
Building dependency tree... 50%
```

When configuring the environment, you need to select the corresponding product. Please enter "2" to download the code for the corresponding product.

```
Please select your kit type:
1. FNK0108: Freenove Computer Case Kit Mini for Raspberry Pi
2. FNK0113:  Freenove Tower Cooler for Raspberry Pi
Enter your choice (1 or 2): 2
```

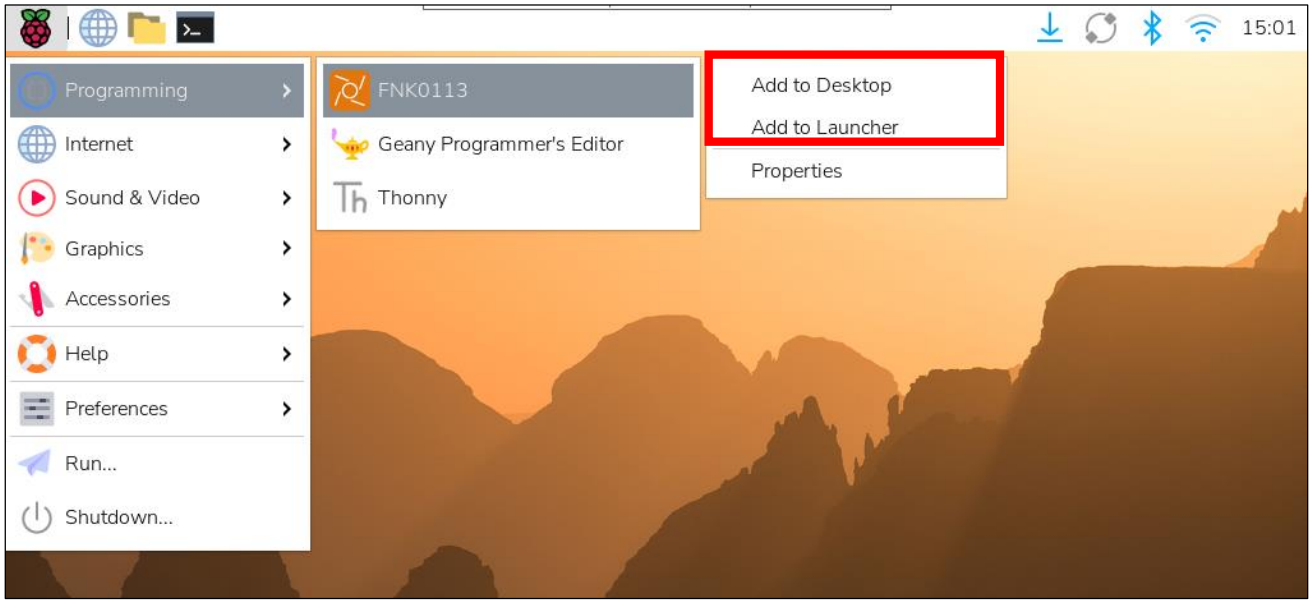


The icon above is a shortcut for the following command.

```
cd Freenove_Tower_Cooler_for_Raspberry_Pi/Code/
python app_ui.py
```

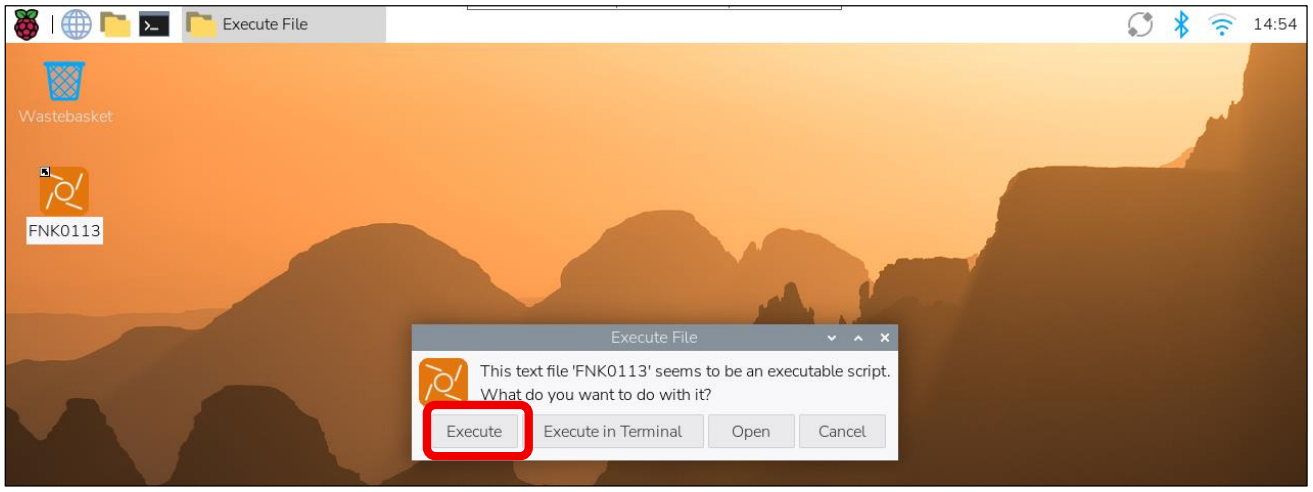
```
pi@raspberrypi:~ $ cd Freenove_Tower_Cooler_for_Raspberry_Pi/Code/
pi@raspberrypi:~/Freenove_Tower_Cooler_for_Raspberry_Pi/Code $ python app_ui.py
```

After completing this step, you will find the application in the "**Programming**" section of the menu. If you accidentally delete the desktop shortcut or wish to add the app to the top launcher, simply right-click "**FNK0113**" and select "**Add to Desktop**"or "**Add to Launcher**".

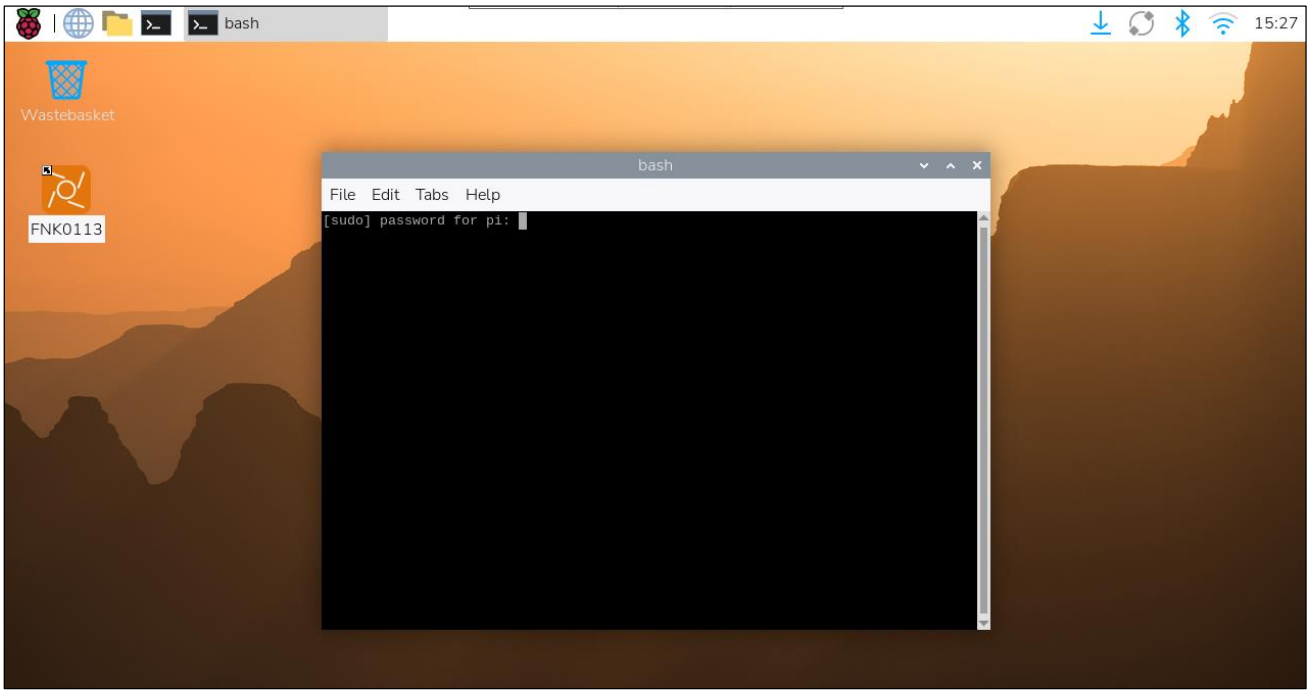


If you are interested in the code implementation, you can explore the files in the Freenove_Tower_Cooler_for_Raspberry_Pi/Code directory.
Should you wish to modify the code, please ensure you back it up first to avoid potential software malfunctions caused by unintended changes.

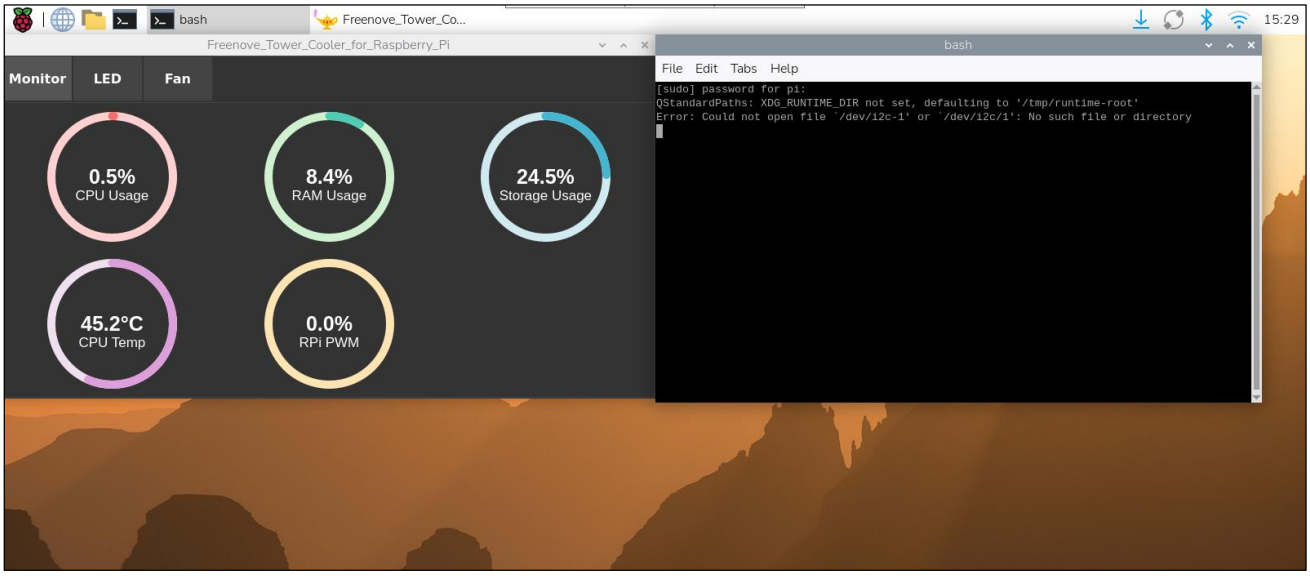
Double click the software with Freenove logo on RPi's desktop, and a window will pop up. Click "**Execute**" to run the program.



Since passwordless sudo is disabled in Trixie, you will also be asked for your password again after reboot. Enter your user password in the pop-up terminal window.



The software interface is shown below. The left window is the main software interface, and the right window displays debugging information.



What's next?

Thank you again for choosing Freenove products.

THANK YOU for participating in this learning experience!

We have reached the end of this Tutorial. If you find errors, omissions, or you have suggestions and/or questions about the Tutorial or component contents of this Kit, please feel free to contact us:
support@freenove.com

We will make every effort to make changes and correct errors as soon as feasibly possible and publish a revised version.

If you want to learn more about Arduino, ESP32, Raspberry Pi Pico W, Raspberry Pi, Smart Cars, Robotics and other interesting products in science and technology, please continue to visit our website. We will continue to launch fun, cost effective, innovative and exciting products.
http://www.freenove.com/

Thank you again for choosing Freenove products.